Sanaul Haque

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Course 608

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IARCO RESEARCH PROPOSAL

Full Legal Name: Dogbe Emmanuel

Institution: Keta Senior High Technical School

Category: Junior

Class/Grade/Year: Gap Year

Country: Ghana

Registered Email Address: emmanueldogbe13@gmail.com

Submission Date: September 20, 2026

Research Topic: Resource-adjusted models for identifying overlooked student academic potential

RESEARCH PROBLEM

College admissions, scholarship programs, and hiring processes often depend heavily on visible signs of ability, such as grades, standardized test scores, awards, and extracurricular achievements. Even though these credentials can be useful in assessing students, they do not entirely define them. A student's ability to earn them can depend greatly on factors outside their control. These factors include the quality of their teachers, access to exam preparation, extracurricular opportunities, reliable electricity and internet, and their family's financial situation. Therefore, two students with similar abilities can end up with very different academic records, and that is just because one had access to more resources than the other.

This proposal considers a more specific question within the broader issue of identifying overlooked talent. Asking whether artificial intelligence can find talented people in general is a broad claim that can be difficult to test. As such, this research examines whether comparing students while taking their access to resources into account can improve on traditional methods that rely mainly on credentials. Accuracy is not the only concern, however. The safety and fairness of such a system are just as important, especially because previous research has shown that predictive models in education can sometimes reproduce existing inequalities rather than reduce them.

The research therefore does not assume that resource-adjusted comparison will necessarily work. The possible outcomes are: identification of students whose abilities are overlooked, provision of little additional value once other factors are considered, and improvement in prediction while creating new fairness concerns. Determining which of these outcomes is most likely is the main purpose of this study.

EXISTING LITERATURE

Machine learning has already been used to identify talent. Zheng et al. developed a semi-supervised neural network trained on data from more than 1,000 secondary school students to identify seven types of talent: academic ability, leadership, and technology skills, earlier than traditional methods [1]. Other research has shown that talented students from disadvantaged backgrounds are often overlooked because of bias in referrals and because they have fewer opportunities to participate in activities that make their abilities visible [2]. A U.S. program targeting underserved students also found that proactive identification increased advanced-course enrollment without lowering average exam performance, suggesting that many of these students were capable of advanced work all along [3].

At the same time, research shows that predictive models can reproduce existing inequalities. Gándara et al. found that a college-success prediction model incorrectly predicted failure for 21% of Hispanic students and 19% of Black students who eventually earned degrees, compared with 12% of White students [4]. Other studies have similarly found unequal error rates in educational and hiring algorithms [5]. Standardized test scores create another concern. SAT

performance is strongly related to family income, even after accounting for previous achievement and parental education [6]. A model that relies heavily on such measures could therefore favor students with greater access to coaching and other resources.

9 These concerns highlight a basic problem: models learn from historical data, and historical data reflects existing inequalities. Resource-adjusted talent identification may still be possible, however; it must be deliberately designed to avoid reproducing those inequalities.

RESEARCH QUESTION

This proposal is guided by a central research question: can a model that explicitly accounts for a student's level of resource access help identify students who are outperforming what their circumstances would predict? Is this a safe and responsible way to identify overlooked talent?

This breaks into two components the research treats as equally important:

- Does resource-adjusted comparison improve identification of high-potential students relative to conventional, credentials-only measures?
- Can this be done without reproducing the demographic bias documented in existing predictive-model research?

METHODOLOGY

Study Design

Rather than proposing a general-purpose system for identifying talent, this proposal outlines a narrower and more testable concept: a residual-performance model. Ranking students by raw academic performance alone favors students with more resources. Therefore, the model first estimates an expected performance level for a student based on measurable, resource-related variables, such as school location, availability of trained teachers, internet or electricity access, and household economic indicators where available. It would then compare each student's actual performance against that expectation. A student who performs significantly above what their resource profile would predict shows a large positive residual, a signal that may indicate strong underlying ability that conventional, unadjusted metrics would undervalue.

The residual score is intended to identify students whose performance exceeds what would be expected from the resources available to them. A consistently high positive residual could serve as a signal for closer, human-reviewed assessment of a student’s potential. The score would be used to identify students for further consideration, not for automatic rankings or admissions decisions. The relationship between residual scores and underlying talent would then be examined empirically to determine whether the signal is meaningful.

Data and Sampling

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A minimal, buildable version of this concept could be tested using West African Senior School Certificate Examination (WASSCE) performance data alongside publicly available school and district-level resource indicators for Ghana. A stratified sample of approximately 50 Senior High Schools across at least three regions (e.g., Volta, Greater Accra, Northern), stratified by locality and infrastructure level, would allow direct comparison between a conventional credentials-only model and a resource-adjusted model on the same population. A simple regression model, comparing predicted performance based on school-level resourcing against actual individual performance, would demonstrate the core mechanism without the complexity of the neural-network approaches used in more advanced talent-prediction research.

Ethical Considerations

Only publicly available, aggregate, non-identifying school-level data will be used in this initial study; no individual student records will be accessed without institutional permission and informed consent. If individual-level data becomes necessary for later validation, ethical approval will be sought from a recognized institutional review board, with informed consent obtained from participating schools and guardians. Data will be stored and analyzed so that no individual student or school is identifiable in published output.

Safeguards

Drawing directly from the bias research reviewed above, any responsible version of this model would need several safeguards built in from the start. First, the model's output should function only as a prompt for human review. The output should be consistent with evidence that bias-mitigation techniques work best paired with ongoing human oversight [5], [9]. Second, the model's accuracy and error rates should be checked separately across regions, school types, and demographic groups, since identical models can perform unevenly across such groups even when overall accuracy looks acceptable [4]. Third, the resource variables used to build the expectation baseline must be chosen carefully, since a poorly chosen baseline could simply encode a different form of the same bias it is meant to correct.

Limitations

At this stage, the proposal is only a conceptual model. The next step would be to develop and test a working prototype using real WASSCE results alongside data on school resources. The residual-scoring approach has not yet been tested against actual outcomes in the Ghanaian context, so its practical accuracy cannot be assumed. Its foundation, however, comes from established statistical methods.

One of the main challenges would be obtaining reliable information about the resources available to individual students. Ghana’s current data infrastructure may not provide enough detailed, student-level information for this purpose. As a result, a real implementation may need to use school- or district-level measures as proxies for individual circumstances. These proxies would introduce some uncertainty, which the research would need to identify and account for when evaluating the model.

RESEARCH TOPIC

Existing research shows both real promise and real danger in using data-driven methods to identify overlooked student talent. Machine-learning models have demonstrated an ability to detect diverse forms of talent beyond simple grades, and targeted identification efforts have shown that underserved students are frequently ready for advanced opportunity long before anyone notices. At the same time, a substantial body of research shows that predictive models in education can just as easily reproduce the inequities they are meant to solve.

I consider this topic urgent because most current identification tools, in Ghana and elsewhere, still rely on the credentialed signals this proposal argues are systematically unequal. A resource-adjusted approach reframes the problem: it asks who is outperforming what their circumstances would predict. That framing has not, to our knowledge, been tested specifically in the Ghanaian WASSCE context. This proposal's residual-performance approach narrows the ambition of talent-identification technology to something testable and honest about its own limitations, treating algorithmic output as a signal for human attention. The proposed next step is to build and evaluate this model using real, ethically sourced Ghanaian examination and school-resourcing data, and to report openly whether resource adjustment helps, does little, or introduces new problems of its own.

QUALITY OF WRITING

This proposal uses plain, precise language throughout and avoids unnecessary jargon, aiming for clarity for a multidisciplinary review audience. Claims are qualified: the proposal treats resource-adjusted identification as a hypothesis to be tested, and distinguishes clearly between what existing literature supports and what remains to be validated.

REFERENCES

[1] X. Zheng et al., "Predicting multi-type talented students in secondary school using semi-supervised machine learning," arXiv, 2025. [Online]. Available: https://arxiv.org/abs/2509.00863

[2] U.S. Department of Education, Institute of Education Sciences, "Underserved gifted and talented students: A review of barriers and a proposed model of talent development," ERIC. [Online]. Available: https://files.eric.ed.gov/fulltext/ED566253.pdf

[3] Fast Company, "Underserved students interested in STEM are often overlooked," 2025. [Online]. Available: https://www.fastcompany.com/91369396/underserved-students-interested-in-stem-are-often-overlooked

[4] D. Gándara, H. Anahideh, M. P. Ison, and L. Picchiarini, "Inside the black box: Detecting and mitigating algorithmic bias across racialized groups in college student-success prediction," Journal of Data Science, 2024. doi: 10.1177/23328584241258741

[5] YIP Institute, "Ensuring fairness in AI: Addressing algorithmic bias in education and hiring," 2025. [Online]. Available: https://yipinstitute.org/capstone/ensuring-fairness-in-ai-addressing-algorithmic-bias

[6] E. J. Dixon-Román, H. T. Everson, and J. J. McArdle, "Race, poverty and SAT scores: Modeling the influences of family income on Black and White high school students' SAT performance," Teachers College Record, vol. 115, no. 4, 2013. doi: 10.1177/016146811311500406

[7] Renzulli Center for Creativity, Gifted Education, and Talent Development, "Finding talent: An overview of research about best practices in identification," Education Sciences, vol. 14, no. 9, 2024. doi: 10.3390/educsci14091027

[8] A. Johnson, "Bridging the gap for underserved populations: Personalized AI solutions for college access and learning support," New Directions for Higher Education, 2024. doi: 10.1002/he.20511

[9] Schiller International University, "Risks of AI algorithmic bias in higher education," 2025. [Online]. Available: https://www.schiller.edu/blog/risks-of-ai-algorithmic-bias-in-higher-education/